Delay insensitive RSFQ circuits with zero static power dissipation

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Abstract - Total power dissipation in RSFQ circuits consists of two parts, dynamic and static. Dynamic power is dissipated in Josephson junctions performing useful logical and data transmission operations. This dissipation is fundamental and proportional to the data rate (at 4 K, of the order of 10^{-18} Joule per bit). Static power is dissipated in resistors used by RSFQ circuits to distribute dc bias current between Josephson junctions. This part of dissipation is not intrinsic to RSFQ circuits and in principle can be eliminated. The goal of this work is to show that Delay Insensitive (DI) RSFQ primitives can be modified so that resistors are no longer required in the dc power supply distribution network, so that the on-chip static power dissipation is absent. In this report we present the schematics for such primitives, define the class of circuits that allow resistor-free current distribution network, and formulate the requirements to the design of this network.

I. INTRODUCTION

Low power dissipation of logic circuits is vital for the Petaflops project [1]. RSFQ circuits are considered as the best candidate for this project since they offer lower power dissipation at very high speed. The power dissipation P_t in RSFQ circuits is the sum of the dynamic dissipation $P_d=I_c \cdot \Phi_0 \approx 2 \cdot 10^{-19}$ Joule per Josephson junction switching, which is associated with performing logical operations, and static dissipation P_s of the resistive power distribution network. In typical large scale (few thousand Josephson junctions) RSFQ circuits P_s dominates the total power budget, for example the efficiency $E=P_d/P_t$ of the RSFQ digital autocorrelator [2] was about $2 \cdot 10^{-2}$. For relatively simple circuits the contribution of P_s can be decreased at the expense of relative bias voltage margin degradation [2] (see Fig. 1).

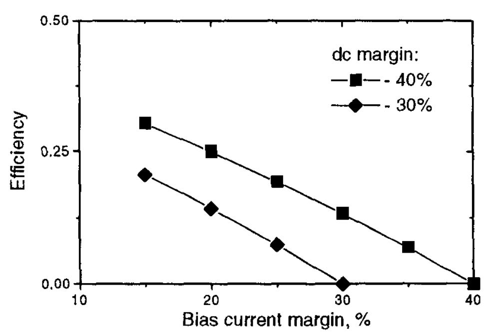

Figure 1: Efficiency of RSFQ circuit as a function of bias margin degradation

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In particular, the experiment [2] has shown that an RSFQ XOR gate can operate at a frequency of 15 GHz with E=0.25 for an input sequence 1⊕1=0 and bias margin degradation from 60% to 20%. For practical circuits with typical bias margins around 30%, margin degradation by more than 10% is hardly desirable. As a result one could expect $E \approx 0.125$ assuming that the circuit operates 100% of the time. For a general purpose processor this estimate may be significantly lower since the workload cannot be distributed so that all subsystems operate 100% of the time.

A universal set of asynchronous Delay Insensitive (DI) RSFQ primitives, suggested in 1997 by the researchers from Intel, SUNY-Stony Brook, and University of Texas in Austin [3], was initially addressing such issues as implementation of asynchronous logic gates in RSFQ technology, their complexity and speed. It was shown that DI RSFQ primitives are at least comparable in area, speed, and parameter margins to existing implementations of Boolean primitives in RSFQ, in sharp contrast to the situation in CMOS. As a building block of asynchronous circuits, they have an important advantage over existing clocked RSFQ logic elements on that they require no minimum separation in the arrival times of input events to work correctly.

We have found that the static power dissipation P_s of DI RSFQ gates can be completely eliminated by means of a non-resistive, purely superconductive power distribution network. and Josephson-junction-based phase-compensating bias current injectors. For these gates $P_s \equiv 0$ at the expense of some increase of P_d associated with switching of the phase-compensating Josephson junctions. In a sense, this fact brings an analogy with CMOS with its negligible static power dissipation. From the Petaflops project point of view, zero static power dissipation might be the most important feature of DI RSFQ gates. This fact, combined with advantages arising from the asynchronous nature of DI RSFQ (modularity, average speed of operation as compared with the worst case speed of clocked circuits) might become an important contribution to the Petaflops project.

The objective of this report is twofold. In Section II we present arguments why it is impossible for classical RSFQ circuits to avoid using resistive bias distribution networks and give estimates on the efficiency of complex parallel circuits. In section III we show how DI RSFQ primitives can be modified to allow purely inductive power distribution network and formulate design requirements for circuits with zero static power dissipation.

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II. Bias network in classical RSFQ circuits

A. Resistive bias network.

Classical RSFQ data representation assumes that a logical "1" corresponds to an SFQ pulse in a "clock window", and a logical "0" to an absence of such a pulse. Fig. 2a presents a two parallel segments of a classical Josephson Transmission Line (JTL). The upper JTL (input *a*) transmits logical "0"s, the lower JTL (input *b*) "transmits" logical "1"s.

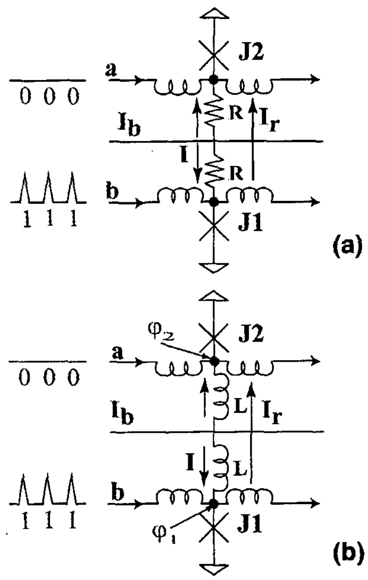

Figure 2: Bias of classical RSFQ JTL: (a) resistive (b) inductive

For the resistively biased JTL, the circuit bias current I_b is divided equally between both junctions *J1*, *J2* so that the junction bias current $I=Ib\cdot R_\Sigma/R$ where R is the value of bias resistor, and, in our example, $R_\Sigma=R/2$. The current I is usually selected as $I\approx 0.7\cdot I_c$, where I_c is the critical current of a Josephson junction. In addition to current I, there exists a "redistribution" current $I_r=V/2R$ flowing into "0" transmitting JTL, which is caused by a voltage drop across the switching junction $J1$: $V=f\cdot\Phi_0$, where f is the frequency of pulses at *J1*, and $\Phi_0=2.07\cdot 10^{-15}\ Wb$, the magnetic flux quantum. For N parallel JTLs each transmitting "1"s except one, the redistribution current into the later JTL is

$$I_r\approx f\cdot\Phi_0/R, \tag{1}$$

assuming $N>>1$, for example $N>10$. To ensure the correct circuit operation for *any* input data, one should select such a value of R that $I_r<<I$. Introducing the ratio $x=I_r/I$, and using expressions for dynamic and static power dissipation

$$P_d=f\cdot\Phi\cdot_0 I,\ P_s=I\cdot R, \tag{2}$$

one gets the following estimate on efficiency

$$E=x/(1+x) \tag{3}$$

For typical RSFQ circuits, the margin on bias current I is about ±30%, so that the value $x=5\text{-}10\%$ seems like a reasonable compromise. If the estimate (3) is valid for arbitrary complex RSFQ circuits, one might expect the best case efficiency of these circuits in the 4.8-9% range.

B. Inductive bias network

A natural question arises if relatively low efficiency of resistively biased classical RSFQ circuits can be overcome by using inductive power distribution of the sort depicted in Fig. 2b. Here, the bias resistors are replaced with corresponding inductors L, so that the bias current I_b is split equally between junctions $J1$, $J2$: $I=I_bL_\Sigma/L$ (for this particular example, $L_\Sigma=L/2$). Now, unlike the resistive bias, the redistribution current $I_r=(\varphi_1-\varphi_2)/2L$ depends on the difference of phases φ_1, φ_2 across junctions *J1*, *J2* respectively. If lower JTL transmits "1" and the upper one – "0"s then after approximately $M=2\cdot L\cdot(I_c-I)/\Phi_0$ pulses the current I_r will be large enough to cause switching of *J2*, which is an error, and cannot be tolerated.

As one can see, the reason why inductive biasing cannot be used in classical RSFQ is the data-dependent phase accumulation at different current injection points of a circuit. In other words, classical RSFQ data representation excludes inductive biasing schemes.

III. Inductive bias network in DI RSFQ circuits

A. Data transmission

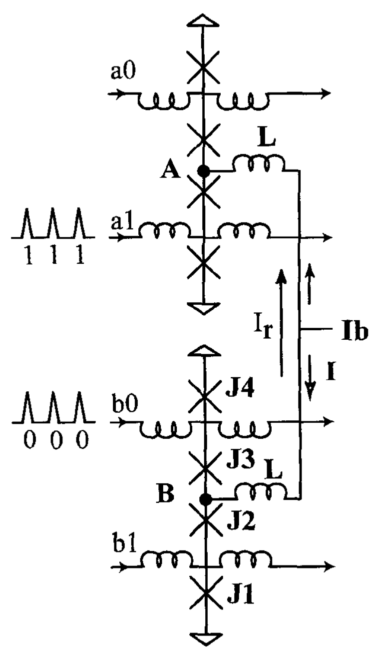

Figure 3: Inductive bias of dual-rail JTL

Dual-rail data encoding enables clock free data transfer [4], [5]. For this type of encoding, a pair of (true-and false-) data lines carry 1-bit binary information. The propagation of a pulse on the true-line or the false-line represents logical "1" or "0" respectively. From the "inductive biasing" point of view, the most important feature of the dual-rail data encoding is that both logical "0" and "1" are transmitted using SFQ pulses. This fact can be used to equalize phase accumulation at different points of the circuit, which, in turn, opens the way for using the inductive biasing.

Fig. 3 introduces an inductive biasing scheme for dual-rail data transmission. The circuit depicted serves the same purpose as those from Fig. 2 - it propagates 2-bit digital signals *a* and *b*. Each signal is represented by two JTLs. For example, the "0"s of signal *b* (the lower part of the circuit) propagate through junction *J4*, "1"s propagate through junction *J1*. The critical currents of the junctions *J2*, *J3* are somewhat smaller then those of *J1*, *J4*. As a consequence, whenever *J1* switches, *J3* switches too, compensating the phase difference in the loop *J1*, *J2*, *J3*, *J4*. Similarly, the switching of *J4* leads to the switching of *J2*. Later on we will refer to the junctions, serving the function similar to that of *J2*, *J3*, as *phase-compensating junctions*. The described dual-rail circuit (which is, by the way, similar to a confluence buffer [7]) ensures that the phase at the current injection point *B* increases by 2π everytime signal *b* arrives, disregarding whether it is "0" or "1". In what follows, we will refer to circuits of that type as *phase-compensating circuits*.

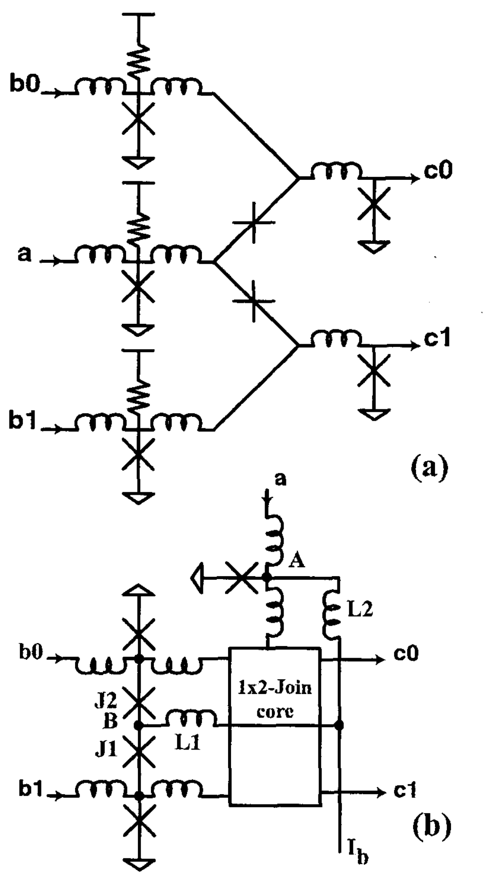

Figure 4: 1x2-Join: (a) with resistive bias, (b) phase-compensating

If the signals *a* and *b* have the same rate, the maximum phase difference between current injection points A and B will not exceed 2π, so that the resulting redistribution current $I_r < \Phi_0/2L$. Its value can be made arbitrary small by increasing the bias inductance *L*.

B. Data storage and processing

For dual-rail circuits, efficient delay insensitive primitives have been proposed [3] (see also report [6] at this conference). A 1x2-Join serves as a destructive readout register with a dual-rail data input/output and a request input. Note that the order of a data pulse and a request pulse is arbitrary. Another primitive, the 2x2-Join serves as a logic gate. It has two dual-rail inputs and four outputs one of which produces a pulse in response to four possible input patterns ("00", "01", "10", "11"). It can implement any 2-input binary operation by merging its outputs with confluence buffers [6].

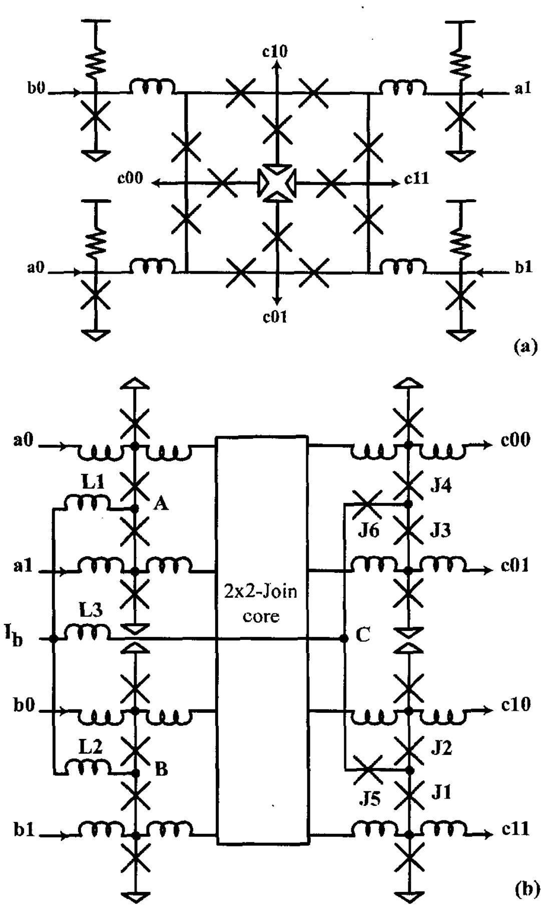

Figure 5: 2x2-Join: (a) with resistive bias (output JTL not shown), (b) phase-compensating

Both circuits can be made phase-compensating. Fig. 4a presents the original schematic of a 1×2-Join (for details see

[3] or [6]), given for the sake of comparison, and Fig. 4b its phase-compensated analog. The part of the circuit, irrelevant for the further discussion, is enclosed into a box titled "1×2-Join core". The proper bias to the circuits is provided by an inductive tree *L1*, *L2*. The dual-rail data input *b* is phase-compensated by adding junctions *J1*, *J2* and injecting bias current distributed by inductance *L1* in point *B*.

Please note that there is no need to phase-compensate single-rail read-out input *a*, since the phase accumulation at point A has the same rate as that at point *B*.

Fig. 5 presents a phase-compensating version of a 2×2-Join. The bias current I_b is injected into points *A*, *B*, and *C* using inductive tree *L1*, *L2*, *L3*. Dual-rail inputs *a* and *b* are phase compensated in a way similar to the data input of the 1×2-Join. The four-rail output *c* is phase-compensated by the Josephson junction tree *J1*, *J2*, ..., *J6*. The example below illustrates the operation of this tree. Let us assume that the input to the circuit is $a{=}b{=}0$. As soon as *a0* and *b0* pulses are present, the output *c00* produces a pulse, which causes junction *J3* to switch, which, in turn, causes junction *J5* to switch. As a result, the phase at the point *C* increases by 2π, i.e. this current injection point is phase-compensated. Similarly, the input $a{=}1$, $b{=}0$, produces a pulse at the output *c10* and causes switching of junctions *J1* and *J6*.

C. Requirements on inductively biased circuits.

For complex circuits consisting of many cells the following requirements should be met to allow inductive dc power distribution network:

1. All cells must be phase-compensating, i.e. for one single operation the phase increment at the cell's bias current injection point is $2\pi{\cdot}n$, where n is constant throughout the circuit. Typically, n would be equal to one.
2. For any single input operand or a set of operands, every cell in a circuit must perform the same number of operations. Typically, this number would be equal to one.

In the case of DI RSFQ primitives, *all* primitives can be made phase-compensating (the trivial cases of 1×1-Join, Merger, and Fork are not considered in this paper). The DI RSFQ primitives form a *universal* set of circuit primitives such that any circuit is realizable as a network of the primitives [8], which means the first requirement does not impose serious restrictions.

The second requirement does not impose serious restrictions either. In fact, all synchronous circuits operating at a single clock rate (single-rate circuits) or their asynchronous counterparts satisfy this requirement. Multiple-rate circuits can be designed using several independent inductive power distribution networks (one for every clock rate) if the number of different clock rates is not too large.

D. Efficiency

There is no static power dissipation in inductively biased circuits based on DI RSFQ primitives. Nevertheless, about a half of Josephson junctions in these circuits are phase-compensating, and do not perform "useful" computations. Thus, the circuit efficiency can be crudely estimated as $E{=}50\%$.

IV. Conclusion

We presented arguments why it is impossible for classical RSFQ circuits to avoid using resistive bias distribution networks and gave estimates on values of bias resistor for complex parallel circuits. We also demonstrated how DI RSFQ primitives can be modified to allow purely inductive power distribution networks with zero static power dissipation, and formulated design requirements for such circuits.

Acknowledgment

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